Sr.No.4106

Master in Computer Application (MCA) (Two Years Programme) - 1st Sem. (Credit Based Evaluation & Grading System) (2024-26)

(2224)

Paper: CSL-4050 Design and Analysis of Algorithms

Time Allowed: 3hrs. **Max. Marks: 70**

Note: Attempt five questions in total by selecting at least one question from each section. The fifth question can be selected from any section.

Section A

1. Explain the procedure of the selection sort algorithm while sorting the elements 70, 30, 20, 50, 60, 10, and 40. Also, write the algorithm of the selection sort. (14)
2. Write in detail about the procedure of designing an efficient algorithm. (14)

Section B

3. (a) Obtain Huffman's encoding for the following data (10)

Character	A	B	C	D	E	F
Frequency	39	10	9	25	7	3

(b) Write the differences between Prim's method and Kruskal method. (4)

4. Solve the following 0/1 knapsack problem using dynamic programming. There are five items whose weights and values are given in the following arrays.
   Weight w[ ] = {1, 2, 5, 6, 7}
   Value v[ ]= {1, 6, 18, 22, 28}
   Find out the optimal knapsack items for the weight capacity of 11 units. (14)

Section C

5. Consider a set S= {5, 10, 12, 13, 15, 18} and d =30. Solve it to obtain the sum of subset. Draw a portion of the state space tree for solving this sum of subset problem. (14)
6. Solve the assignment problem for the following jobs using branch and bound. (14)

	Job 1	Job 2	Job 3	Job 4
Person A	20	1	7	8
Person B	7	4	3	6
Person C	5	8	1	8
Person D	7	6	20	4

Section D

7. Write a note on the following: (3.5 x 4)
   (a) P class problems
   (b) NP class
   (c) NP hard
   (d) NP complete
8. Explain Breadth first and Depth first techniques with the help of examples. Also, write their algorithms. (14)

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